

L1 Technical Discussion- Lalit kumar-8010767541- Camera
Engineer- 15-May-26 @ 1.00 PM- KPIT Technologies

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📄 👍 💬 ↗️ ⋮ 📖 Sources

Write everything in details camera hal and camera kernel flow

You're giving feedback on a new version of ChatGPT.
Which response do you prefer? Responses may take a moment to load.

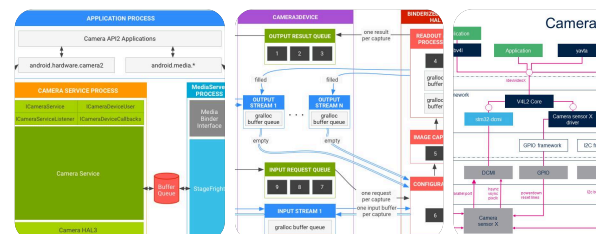


Response 1

```
Camera App
↓
Android Framework
↓
Camera Service
↓
Camera HAL
↓
Kernel Driver
↓
Sensor Hardware
```



Response 2



```
Camera App
↓
Camera2 API
↓
Camera Framework
(CameraManager / CameraDevice / CaptureSession)
↓
CameraService
```

```

+-----+
| Camera App |
| (WhatsApp/Instagram) |
+-----+
      |
      v
+-----+
| Camera2 API |
| Android Framework |
+-----+
      |
      v
+-----+
| CameraService |
| (system_server) |
+-----+
      |
      v
+-----+
| Camera HAL (Vendor) |
| libcamera.so |
+-----+
      |
      v
+-----+
| Kernel Driver |
| V4L2 / I2C / CSI |
+-----+
      |
      v
+-----+
| Camera Sensor HW |
+-----+

```

```

Boot
↓
camera_driver.ko loaded
↓
probe() called
↓
sensor detected over I2C

```

```

↓ Binder IPC
Camera HAL (vendor)
↓
Kernel Driver (V4L2)
↓
Sensor + ISP + CSI + I2C
↓
Image captured

```

```

CameraManager
openCamera()

```

```

</> C++

camera_module->open()

```

</> C++



```
static int camera_probe(struct i2c_client *client)
{
    printk("Camera sensor found");
}
```

Write register 0x0000

Read value

Expected: 0x5640



Sensor found



1. regulator_enable() → power
2. clk_prepare_enable() → clock
3. gpio reset
4. gpio pwn



Sensor not detected



Sensor probe fails.
I2C read timeout.
No camera detected.



</> C++

i2c_transfer()

</> Java

createCaptureSession()

</> C++

configureStreams()

</> Java

captureRequestBuilder.build()

</> C++

processCaptureRequest()

Video4Linux2



</> C++



```
v4l2_async_register_subdev(sd);
```

There is a camera device



Sensor ---> CSI PHY ---> ISP



4 lanes
1.5 Gbps/lane



Lane count
Clock rate
Pixel format



Frame corruption
No image



VIDIOC_STREAMON



Start sending frames



</> C++

```
ioctl()
```

Exposure register
Gain register
Frame length
Mode select

Sensor
→ CSI lanes
→ ISP
→ scaling
→ denoise
→ color correction
→ YUV output
→ memory

</> C++

```
ISR()
```

</> C++

```
VIDIOC_DQBUF
```

0x0100 = 0x01



Start streaming



Sensor → MIPI CSI → ISP → DMA → RAM



Direct Memory Access



Buffer1
Buffer2
Buffer3



Triple buffering



Sensor writes frame into buffer
Driver marks buffer ready
HAL consumes
Reuse buffer



</> C++

```
processCaptureResult()
```

```
probe()
open()
s_stream()
queue buffers
interrupt
dequeue
stop
close
```

</> C++

```
static int sensor_probe(...)
{
    power_on();
    read_chip_id();
    init_registers();
}
```

</> C++

```
v4l2_open()
```

Exposure setting
Gain setting
Focus



Write exposure register



Millions of pixels/sec



</> C++



VIDIOC_QUERYCAP
VIDIOC_S_FMT
VIDIOC_REQBUFS
VIDIOC_QBUF
VIDIOC_DQBUF
VIDIOC_STREAMON
VIDIOC_STREAMOFF

Frame done!



</> C++



irq_handler()

</> C++

sensor_write(STREAM_ON)

</> C++

VIDIOC_QBUF

</> C++

frame_done_irq()

</> C++

VIDIOC_DQBUF

</> C++

s_stream(0)

Framework ↔ HAL ↔ Kernel



Give me preview at 1080p



Configure sensor
Set ISP
Allocate buffers
Start stream



camera_provider
camera_device
camera3_device_ops



</> C++

get_number_of_cameras()



2 cameras



</> C++

HAL_PIXEL_FORMAT_YCbCr_420_888

</> C++



```
camera_device_open()
```

sensor handles



ISP handles

buffers

threads

Capture this frame



Exposure = X

ISO = Y

Focus = Z

</> C++



```
process_capture_request()
```

Program sensor



Queue buffer

Wait for frame

Frame data



Metadata

Timestamp

AE/AF values

```
Black level correction
Dead pixel correction
Lens shading correction
Demosaic
White balance
Color correction
Noise reduction
Gamma correction
Sharpening
YUV conversion
Scaling
```

R G

G B

Framework request



process_capture_request()



Kernel streams



Frame ready



process_capture_result()



Framework receives image



Java



```
CameraManager.openCamera()
```

Binder IPC



Permission

Camera availability



vendor/lib64/hw/camera.xxx.so



Bash

```
logcat | grep Camera  
dmesg
```

Need preview:
1920x1080 YUV



Sensor mode
ISP output
Buffers



Preview request
Preview request
Preview request



VIDIOC_QBUF
STREAMON



Light
↓
Lens
↓
Sensor
↓
RAW Bayer



RAW Bayer
↓
Demosaic
↓
Noise Reduction
↓
White Balance
↓
Color Correction
↓



</> C++

VIDIOC_STREAMON
VIDIOC_QBUF
VIDIOC_DQBUF

Tone Mapping



Sharpening



YUV/JPEG

R G

G B



Exposure time

Sensor gain



VCM driver



Sensor



CSIPHY



CSID



ISPIF



VFE



CPP



HAL

Preview running
User presses shutter



High resolution request



12 MP RAW



JPEG



Talks to hardware directly



Talks to Android framework



App → Camera2 API → CameraService
→ HAL open
→ Configure streams
→ Kernel STREAMON
→ Sensor sends frames
→ HAL returns buffers



I prefer this response

Exposure
Gain
Focus



Frame drops
Tearing
Latency



Permission?



adb logcat | grep Camera



process_capture_request



⌕ Bash



dmesg | **grep** camera

I2C probe success?



STREAMON success?



CSI errors?



APP asks
Framework manages
HAL translates
Kernel controls
Sensor captures
ISP processes
Buffer returns



App → Framework → HAL → Kernel → Sensor
Sensor → ISP → Kernel → HAL → App



I prefer this response